

Engineering Innovation Challenge 2023

Flood-Proof Roads

Team U-9

Jason Abi Chebli

Phan Tuan Kiet Le

William Wang

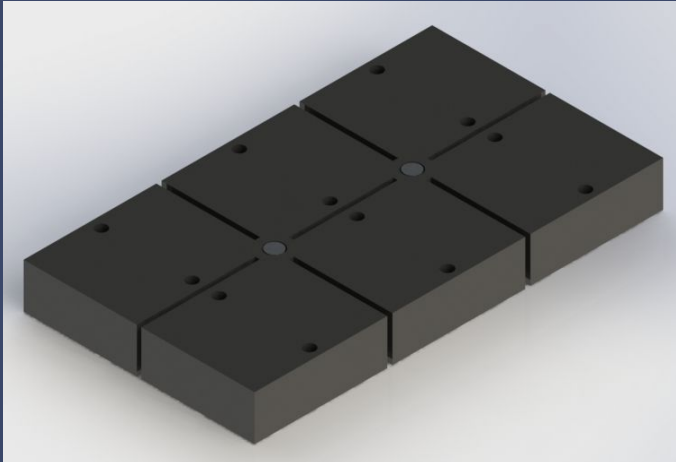
Research Problem



An ambulance is seen stuck in flood water near Changi fire station in Singapore (Video: Reddit / Penguini)

Design and Methodology

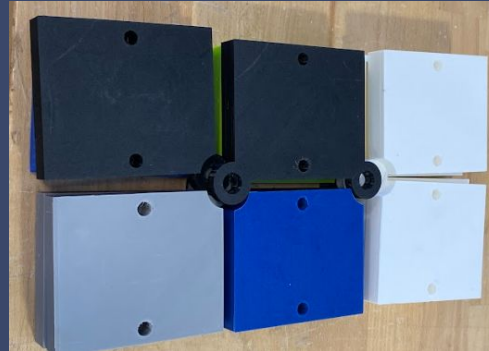
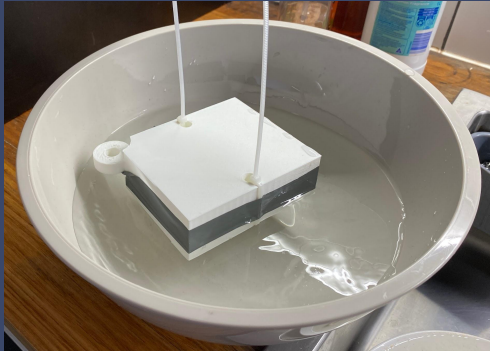
a) Scaling



```
1 clc; clear all; close all;
2
3 %% To Scale Calculations
4
5 % Constant variables we know
6 rho = 1000; %density in kg/m^3
7 g = 9.81; %gravity in m/s^2
8 mass = 30000; %mass of the heaviest vehicle kg
9 width = 3.2; %width of the road (standard two lane road is 3.2m, expecting the pontoon to be at least half)
10
11 % Determine the height of the pontoon
12 length = 0.1; %width % need to determine the optimal length
13 height = mass./(rho*length*width); %the height is dependent on the length
14
15 figure()
16 plot (length,height)
17 xlabel('length (m)')
18 ylabel('Height (m)')
19 title('Dimensions for the pontoon (To Scale)')
20
21 %% Determine small scale (ss)
22 widthss = 0.1; % Assume we want the width of one pontoon to be 10cm (0.1m)
23 scale = width/widthss; %this is the scaling factor needed to apply to all the variables to make it small scale
24 lengthss = length/scale; % need to determine the optimal length
25 heightss = height/scale; %the height is dependent on the length
26 massss = rho.*lengthss.*heightss.*widthss; %mass of the small scale
27
28 figure()
```

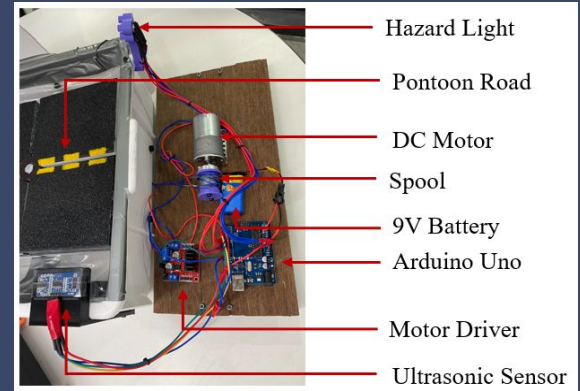
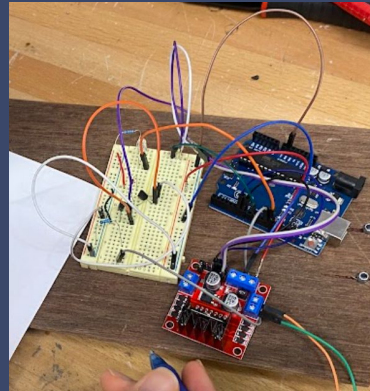
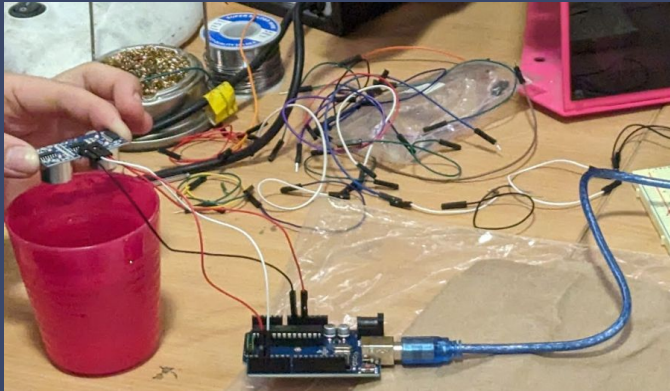
Design and Methodology

b) Pontoon Road



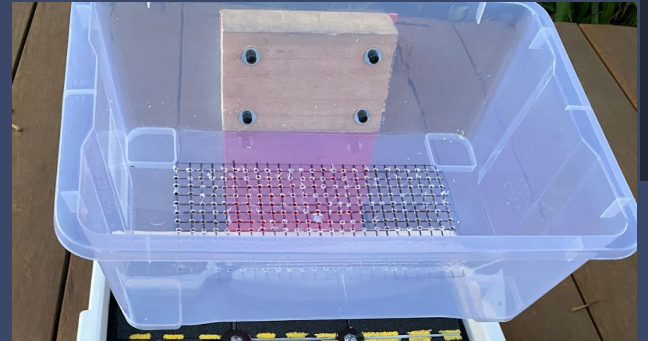
Design and Methodology

c) Deployment System

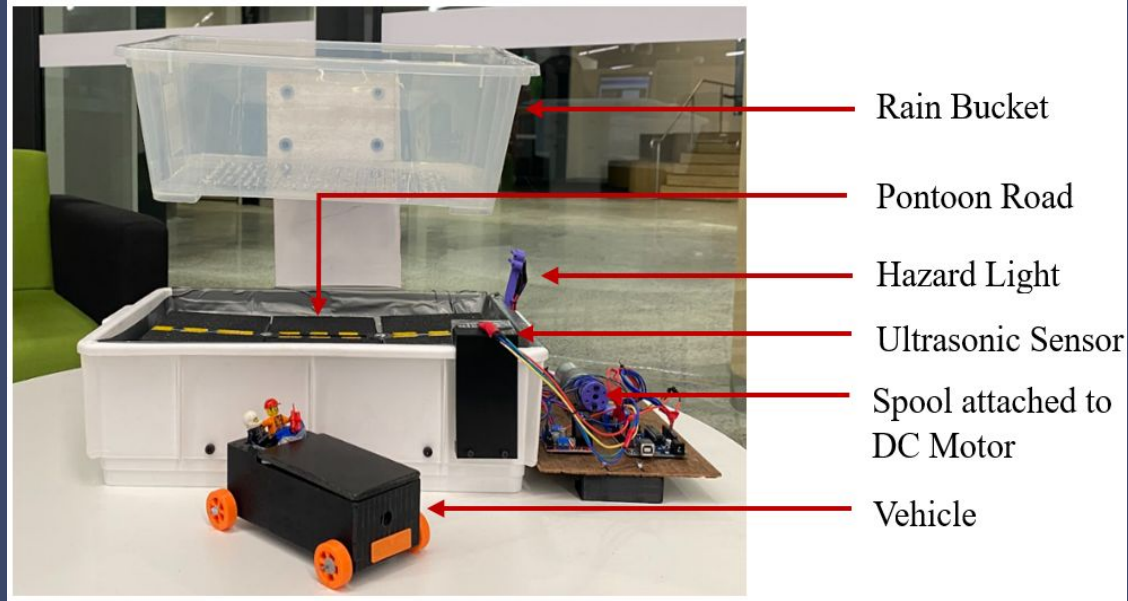


Design and Methodology

d) Simulating Environment



Final Prototype

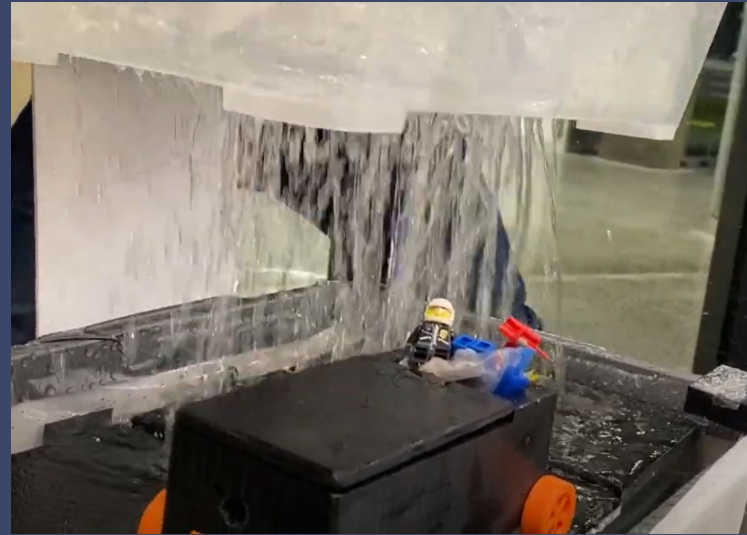


Testing

The prototype underwent rigorous evaluation in two main testing scenarios:



1. Dry Conditions



2. Flash Flooding Conditions

Evaluation

- ❑ Resilient solution with validated scalability and adaptability, enables safe commuting on flood-prone roads
- ❑ The prototype met the engineering goals:

Buoyancy and Stability

Structural Integrity

Adaptability to Dry and Flood Conditions

Appropriate Material Selection

- ❑ Further testing and refinement are necessary:

Different Pontoon Designs

Different Rainfall Conditions

Yielding and Life Cycle Testing
of the Pontoon

Assessing Faster Moving Cars

Scalable Materials